Omar A. Beg, Ph.D., PMP

Associate Professor and Chair
Computer Science & Electrical Engineering
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EDUCATION & CERTIFICATIONS

Ph.D. Major: Electrical Engineering, University of Texas at Arlington, 2014–2017.
Dissertation: Formal Verification of DC Distribution Networks. (Ph.D. CGPA: 4.0/4.0)

M.S. Major: Electrical Engineering, National University of Sciences and Technology, 2004–2006.
Minor: Control Engineering.
Thesis: Parameter Estimation of Linear Time-Invariant Systems. (CGPA: 3.5/4.0)

B.S. Major: Electrical Engineering, National University of Science and Technology, 1994–1998.
Senior Design Project: Design and Development of Heartbeat Monitor.

ABET IDEAL Scholar Certificate, ABET Institute for the Development of Excellence in Assessment Leadership (IDEAL), ABET Headquarters, USA, 2024.

Certificate in Effective College Instructions, The American Council on Education (ACE) and The Association of College and University Educators (ACUE), USA. 2021.

Project Management Professional (PMP) Certification, Project Management Institute, USA, 2013.

AWARDS

UTPB President's Research Award, Conferred by UT Permian Basin President, 2022.
Recognition: Outstanding research performance among the university faculty.

College of Engineering Outstanding Faculty in Research Award, Conferred by the Dean, 2022.
Recognition: Outstanding research performance among the college faculty.

Arlen and Betty Edgar Endowed Fellow in Engineering, Approved by UT System, 2022–Present.
Recognition: In recognition of outstanding research and overall exemplary performance.

Senior IEEE Member, by Institute of Electrical and Electronics Engineers (IEEE), 2022.
Recognition: Honor bestowed to those who made significant contributions to the profession.

Graduate Research Fellowship, by US Air Force Research Lab, Summer 2015.
Program: Office of Scientific Research Summer Faculty Fellowship Program, Rome, NY.

GRANTS SECURED (Around $9 Million)

Grant-1: "Enabling Reasoning, Secure, and Resilient Autonomy for Heterogenous Multi-Robot Fleet in Nuclear Domain", by US Nuclear Regulatory Commission as a Co-PI in collaboration with University of Connecticut, in January 2025 (3 years, **$500,000**).

Grant-2: "Permian Basin Success Initiative (Project IDEA)", by US Department of Education as a Co-PI, in October 2023 (5 years, **$3 Million**).

Grant-3: "Educational Outreach Program – UT Permian Basin", by US Army Research Office under Army Education Outreach Program as a Co-PI, in May 2022 (3 years, **$33,000**).

Grant-4: "Strengthening the STEM Pipeline – UT Permian Basin", by US Department of

Education as a PI, in October 2021 (5 years, **$4.9 Million**).

Grant-5: "Texas Reskilling Supporting Fund Grant", by Texas Higher Education Coordination Board as a PI, in April 2021 (1 year, **$112,500**).

Grant-6: "Research and Education in Artificial Intelligence for Control and Cyber-Security of IoT-Based Power Systems", by US Department of Defense as a PI, in August 2021 (3 years, **$300,000**).

Grant-7: "Resilient and Cyber-Secure Smart Electrical Systems", by UT Systems under Rising STARs Program as a PI, in September 2019 (3 years, **$300,000**).

FACULTY POSITIONS

Associate Professor of Electrical Engineering, Department of Computer Science and Electrical Engineering, The University of Texas Permian Basin, August 2025–**Present**.

Assistant Professor of Electrical Engineering, Department of Electrical Engineering, The University of Texas Permian Basin, August 2019–August 2025.

Assistant Professor of Research/Faculty Associate Researcher, Department of Electrical Engineering/UT Arlington Research Institute, The University of Texas at Arlington, August 2017–August 2019.

Instructor of Electrical Engineering, Department of Electrical and Power Engineering, The University of Sciences and Technology, Pakistan, August 2006–August 2008.

ADMINISTRATIVE AND TECHNICAL POSITIONS

Department Chair, Department of Computer Science and Electrical Engineering, The University of Texas Permian Basin, August 2025–**Present**.

Interim Department Chair, Department of Computer Science, The University of Texas Permian Basin, August 2024–August 2025.

Department Coordinator, Department of Electrical Engineering, The University of Texas Permian Basin, May 2021–August 2025.

Project Director HSI STEM Project, Office of Research and Sponsored Programs, The University of Texas Permian Basin, October 2021–Present.

Activity Director Project IDEA, Office of Research and Sponsored Programs, The University of Texas Permian Basin, October 2023–Present.

Senior Electrical Engineer and R&D Project Manager, Pakistan, 2008–2013.

Electrical Engineer and Training Officer, Pakistan, 1998–2004.

TEACHING EXPERIENCE

The University of Texas Permian Basin:

I can teach a wide variety of courses pertaining to Electrical and Computer Engineering. The following new courses/labs were prepared/developed and delivered in line with the ABET guidelines:

Signals and Systems, Control Systems, Digital Circuit Design and lab, Electromagnetic

Fields, Fundamentals of Circuit Analysis lab, Fundamentals of Circuits Analysis, Electronic Circuit Analysis lab, Foundations of Electrical Engineering, Electric Power Systems lab, Power Electronics, Design Methodology in Electrical Engineering.

Developed the laboratory manuals for various labs and conducted the labs for the first time at The University of Texas Permian Basin for the new electrical engineering program.

Actively participated to prepare the course description of a new course, titled as, “Foundations of Electrical Engineering” to meet the ABET accreditation requirements.

The National University of Sciences and Technology:

- Taught courses pertaining to control systems, power electronics, and circuit analysis
- Performed ISO audit of the education processes and procedures
- Organized three open house (job fair) sessions for local industry
- Organized an international conference in collaboration with IEEE
- Participated in accreditation of the college from engineering accreditation authority
- Provided mentoring to the electrical engineering students
- Arranged and led the industrial visits for the engineering students

ACADEMIC PROFESSIONAL SERVICE & LEADERSHIP

Prepared the ABET Self-Study Report (SSR) for initial accreditation of BS Electrical Engineering program. Also assisted in preparation of the SSR for BS Chemical Engineering program.

Faculty Senate member since 2023 to represent the faculty of Electrical Engineering department

Formed the Electrical Engineering Advisory Board (EEAB) in coordination with the local industry and alumni to advise on academic matters

I have also mentored undergraduate student research. I have regularly supervised and mentored the undergraduate students since Fall 2019. The local industry (i.e., Priority Power and Endeavor Energy Resources) have hired two of those students based on their research work. Presently, I am mentoring the research of one student supported through Arlen and Betty Edgar Endowed Fellowship.

Since Fall 2019 to Spring 2022, I was the co-advisor of a Ph.D. student at The University of Texas at San Antonio. I was approved as the graduate faculty at UT San Antonio. The candidate has successfully defended the doctoral dissertation and graduated during Spring 2022. The student is now employed in electric power industry.

I have secured grant funding that supported student success, student careers, and student advising since 2021. There were many grant funded positions created to include Assistant Director of Student Success Center, Internship Coordinator at the Career Services, and Academic Advisor at the Advising Center.

Academic adviser of undergraduate students at the Department of Electrical Engineering

Mentor/adviser of IEEE Students’ Chapter at The University of Texas Permian Basin during 2020

Faculty Advisor of the local student chapter of National Society of Black Engineers (NSBE) since 2024

Actively participated during annual summer camps (since 2020) at The University of Texas Permian Basin for middle and high school students. Arranged a robotics summer camp during Summer 2020 for the students to provide them hands-on training on assembling, programming, and controlling the robots.

REASEARCH INTERESTS

Developing novel AI-driven cyber-attack detection techniques inspired by multimodal signal processing.

Cyber-attack detection and resilience in networked cyber-physical systems using formal methods and artificial intelligence.

PUBLICATIONS & CONFERENCES

Book

Omar A. Beg, “Foundations of Electrical Engineering”, ISBN 13: 978-1-60797-013-2, Publisher: Linus Learning, NY, USA.

Journal Publications

1. Tipu, J. , Noon, A. , Arif, M. , Beg, O. , Khushnood, S. , Ahmed, S. , Ahmed, R. , Basit, A. , Khan, Z. , Shaharyar, M. and Khan, A., “Sustainable Carbon Management: Minimizing Carbon Contaminants from Exhaust Gases for a Green and Healthy Environment” in World Journal of Engineering and Technology, vol. 14, pp. 226-249, 2026. doi: 10.4236/wjet.2026.141013.
2. Kaza, S. and Beg, O., “Engineering the Quantum Future: Impact of Electrical Engineering in the Field of Quantum Mechanics,” in World Journal of Engineering and Technology, vol. 14, pp. 207-225, 2026. doi: 10.4236/wjet.2026.141012.
3. Tipu, J. , Arif, M. , Noon, A. , Beg, O. , Elharati, H. , Waseem, E. , Kazmi, E. , Khalil, E. , Khan, U. , Khushnood, S. and Zafar, E., “Design and Development of Sustainable Tidal Energy System for Coastal Regions,” World Journal of Engineering and Technology, vol 13, pp. 1016-1040. November 2025. doi: 10.4236/wjet.2025.134061.
4. Mohamadamin Rajabinezhad, N. Shams, Omar A. Beg, Asad A. Khan, and Shan Zuo, ”Lyapunov-Certified Resilient Secondary Defense Strategies of AC Microgrids Under Exponentially Energy-Unbounded FDI Attacks," IEEE Control Systems Letters, vol. 9, pp. 1213-1218, June 2025.
5. H. Elharati, M. Hlal, M. Prince, and Omar A. Beg, ”Design and Implementation of a UAV Platform for Educational Applications: A Multidisciplinary Engineering Approach,” World Journal of Engineering and Technology, vol. 13, pp. 607-621, 2025.
6. Omar A. Beg, Asad A. Khan, Waqas Ur Rehman, and Ali Hassan, “A Review of AI-Based Cyber-Attack Detection and Mitigation in Microgrids,” Energies, vol. 16, no. 22, November 2023.
7. Asad A. Khan, Omar A. Beg, Yu-Fang Jin, and Sara Ahmed, “An Explainable Intelligent Framework for Anomaly Mitigation in Cyber-Physical Inverter-based Systems,” IEEE Access, vol. 11, pp. 65382-65394, June 2023.
8. Asad Ali Khan and Omar A. Beg, “Cyber Vulnerabilities of Modern Power Systems,” Book Chapter in Power Systems Cybersecurity, Publisher: Springer, February 2023.
9. Xiaodong Yang, Omar A. Beg, Matthew Kenigsberg, and Taylor T. Johnson, “A Framework for Identification and Validation of Affine Hybrid Automata from Input-Output Traces,” ACM Transactions on Cyber-Physical Systems, June 2021.
10. Asad Ali Khan, Omar A. Beg, Miltos Alamaniotis, and Sara Ahmed “Intelligent Anomaly Identification in Cyber-physical Inverter-based Systems,” Elsevier Electric Power Systems

Research, vol. 193, April 2021.

11. Omar A. Beg, Luan V. Nguyen, Taylor T. Johnson, and Ali Davoudi, “Cyber-Physical Anomaly Detection in Microgrids Using Parametric Time-Frequency Logic,” IEEE Access, vol. 9, pp. 20012-20021, January 2021.
12. Shan Zuo, Omar A. Beg, Ali Davoudi, and Frank L. Lewis, “Resilient Networked AC Microgrids Under Unbounded Cyber Attacks,” IEEE Transactions on Smart Grid, vol. 11, no. 5, pp. 3785-3794, September 2020.
13. Omar A. Beg, Luan V. Nguyen, Taylor T. Johnson, and Ali Davoudi, “Signal Temporal Logic-based Attack Detection in DC Microgrids,” IEEE Transactions on Smart Grid, vol. 10, no. 4, pp. 3585-3595, July 2019.
14. Omar A. Beg, Houssam Abbas, Taylor T. Johnson, and Ali Davoudi, “Model Validation of PWM DC-DC Converters,” IEEE Transactions on Industrial Electronics, vol. 64, no. 9, pp. 7049-7059, September 2017.
15. Omar A. Beg, Taylor T. Johnson, and Ali Davoudi, “Detection of False-data Injection Attacks in Cyber-Physical DC Microgrids,” IEEE Transactions on Industrial Informatics, vol. 13, no. 5, pp. 2693-2703, October 2017.
16. Stanley Bak, Omar A. Beg, Sergiy Bogomolov, Taylor T. Johnson, Luan V. Nguyen, and Christian Schilling, “Hybrid Automata: From Verification to Implementation,” International Journal on Software Tools for Technology Transfer, vol. 21, no. 1, pp. 87-104 Feb. 2019.

Conference Publications/Proceedings

1. M. Rajabinezhad, N. Shams, A. A. Khan, O. A. Beg and S. Zuo, "Lyapunov-Certified Resilient Secondary Defense Strategies of AC Microgrids Under Exponentially Energy-Unbounded FDI Attacks," in IEEE Conference on Decision and Control (CDC), Rio de Janeiro, Brazil, Dec. 2025.
2. Rammah Abohtyra, and Omar A. Beg, “Body Fluid Estimation during Standard Ultrafiltration in Chronic Kidney Disease,” in American Control Conference (ACC), Denver, CO, USA, 2025, pp. 4659-4664.
3. J. Ngo, P. Lim, G. Kopp, Omar A. Beg, S. Movahed, and P. Kumar, “WIP: Fostering Effective Teamwork in Undergraduate Research Based Course,” in Conference Proceedings at the Frontiers in Education Conference, Nashville, TN, USA, Nov. 2025.
4. J. Ngo, P. Lim, G. Kopp, Omar A. Beg, S. Movahed, and P. Kumar, “WIP: PicPrompt - a Novel Social Prompt Engineering Game Based Tool for Informal Learning in K-12,” in Conference Proceedings at the Frontiers in Education Conference, Nashville, TN, USA, Nov. 2025.
5. Yichao Wang, Omar A. Beg, Mohamadamin Rajabinezhad, and Shan Zuo, ”Distributed Resilient Control of DC Microgrids Under Polynomially Unbounded FDI Attacks," in Proceedings of IEEE PES Grid Edge Technologies Conference and Exposition (Grid Edge), San Diego, CA, USA, 2025, pp. 1-5.
6. Sohan Gyawali and Omar A. Beg, “Cyber Attacks Detection using Machine Learning in Smart Grid Systems,” in Proceedings of IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), New York, NY, USA, 2022, pp. 1-2.
7. Asad Ali Khan, Omar A. Beg, and Sara Ahmed “Intelligent Anomaly Mitigation in Cyber-Physical Inverter-based Systems,” in Proceedings of IEEE Energy Conversion Congress and Exposition, Vancouver, Canada, Oct. 2021.
8. Omar A. Beg, Ajay P. Yadav, Taylor T. Johnson, and Ali Davoudi, “Formal Online Resiliency Monitoring in Microgrids,” in Proceedings of IEEE Resilience Week, Salt Lake City, ID, USA, 2020, pp. 99-105.

9. Omar A. Beg, Luan V. Nguyen, Ali Davoudi, and Taylor T. Johnson, “Computer-Aided Formal Verification of Power Electronics Circuits,” in Proceedings of IEEE Frontiers in Analog CAD (FAC), Frankfurt, Germany, July 2017.
10. Omar A. Beg, Ali Davoudi, and Taylor T. Johnson, “Reachability Analysis of Transformer-Isolated DC-DC Converters (Benchmark Proposal),” in Proceedings of 4th International Workshop on Applied Verification for Continuous and Hybrid Systems (ARCH 2017), Co-located with the Cyber-Physical Systems Week 2017, Pittsburgh, PA, April 2017.
11. Omar A. Beg, Ali Davoudi, and Taylor T. Johnson, “Charge Pump Phase-Locked Loops and Full Wave Rectifiers for Reachability Analysis (Benchmark Proposal),” in Proceedings of 3rd International Workshop on Applied Verification for Continuous and Hybrid Systems (ARCH 2016), Co-located with the Cyber-Physical Systems Week 2016, Vienna, Austria, April 2016.

Under Review Publications

1. “Factors Influencing the Success of Energy Management Software Tools: A Quantitative and Literature- Based Approach,” Under review in IEEE Access.
2. “Design and Development of Sustainable Tidal Energy System for Coastal Regions,” Under review in a journal.

PRESENTATIONS & TALKS

1. Omar A. Beg, “Cyber-Attack Vulnerabilities of Modern Power Systems," Published in online Power Magazine, Houston, TX, September 2023.
2. Mohsin M. Jamali, Raiel Amesquita, and Omar A. Beg, “Quantum Sensor Array Processing Experiment," Presented at the 2023 IEEE 66th International Midwest Symposium on Circuits and Systems, Phoenix, AZ, August 2023.
3. Nikita Naumov and Omar A. Beg, “Maximum Power Point Tracking and Application of Neural Networks in Photovoltaic Systems,” in the Louisiana State University Shreveport Regional Student Scholars Forum, Shreveport, LA, March 2021.
4. Omar A. Beg, “Scalable Formal Verification of Resilient Converter-dominated MVDC Networks,” in the Office of Naval Research Controls Workshop at the University of Virginia, Blacksburg, VA, 2019.
5. Omar A. Beg, Shankar Abhinav, and Ali Davoudi, “Resilient Microgrid Control,” in the Office of Naval Research Controls Workshop, Arlington, TX, 2018.
6. Omar A. Beg, Ali Davoudi, and Taylor T. Johnson, “Detecting and Mitigating Cyber-Physical Attacks with Invariant Inference and Runtime Assurance,” poster presentation at US Air Force Research Laboratory, Rome, NY, 2015.
7. Omar A. Beg, Ali Davoudi, and Taylor T. Johnson, “Formal Verification of Software-controlled Power Electronics,” in US Air Force Research Laboratory Safe and Secure Systems and Software Symposium, Dayton, OH, 2015.
8. Omar A. Beg, Ali Davoudi, and Taylor T. Johnson, “Computer Aided Formal Verification of Power Electronics Based Cyber-physical Systems,” in Formal Methods in Computer Aided Design, Austin, TX, 2015.

INDUSTRIAL & TECHNICAL EXPERIENCE

Initiated, led, and completed various research and development projects related to power electronics

Led the engineering team to smoothly transition the developed products from R & D prototypes to full production that met the performance, reliability, and cost targets

Documented the systems and subsystems level control requirements from functional specifications and the safety aspects in collaboration with all the stakeholders
Led the team during trials of the prototypes, conducted Failure Mode and Effects Analysis (FMEA) and identified the root causes of the failures, implemented technical solutions, provided design and documentation review through problem solving, resulting in improved prototypes that passed the acceptance trials

Prepared the technical progress reports and presented to higher authorities

Led the technical team to perform preventive and corrective maintenance of electrical equipment

VOLUNTEER & OUTREACH ACTIVITIES

Reviewer of the following technical journals:

IEEE Transactions on Smart Grid
IEEE Transactions on Power Electronics
IEEE Transactions on Power Systems
IEEE Transactions on Energy Conversion
IEEE Transactions on Circuits and Systems - I
IEEE Transactions on Industrial Informatics
IEEE Transactions on Industrial Electronics
IET Control Theory and Applications

Conducted the outreach activities at The University of Texas at Arlington for local schools of the Arlington Independent School District to apprise the school kids about the importance of science, technology, and engineering education and develop their interest in this profession (2016-2017).

PROFESSIONAL MEMBERSHIPS

Institute of Electrical and Electronics Engineers (IEEE) – Senior Member
Project Management Institute (PMI) – Member
American Society of Engineering Education (ASEE) – Member
IEEE Power and Energy Society (PES)
IEEE Technology and Engineering Management Society
IEEE PELS Technical Committee on Design Methodologies
Tau Beta Pi - The Engineering Honor Society